LOCAL STRUCTURE OF ÉTALE ALGEBRAS: THE PROOF AS FORMALISED IN UNIT

Abstract. Let R be a residually discrete local ring, E an étale R-algebra and $\mathfrak{q}$ a decidable prime of E above $\mathfrak{m}$. Then $E[1/s']$ is standard étale for some $s' \notin \mathfrak{q}$. This is the local theorem of the note *Local structure of étale algebras* (`zmt2-revised.tex`). We give its proof in the form that has been checked in Unit. That form simplifies the note at several points: a remainder loop replaces gcds, there is no gcd with X, Section 1 is read in E itself, and the derivative step uses dual numbers. The last section relates the proof to the formal development, where Zariski's Main Theorem is no longer a hypothesis: it is the theorem proved in `sag/zmt-general`. What remains a hypothesis is the structure of finite étale algebras over a discrete field. The same conclusion is proved, and checked, for E finitely presented, flat and unramified, with an ending that replaces the special-fibre comparison and Nakayama's lemma of the note by a flatness transfer theorem.

1. Definitions and statement

All rings are commutative. An R-algebra E is *étale* if it is finitely presented and *formally étale*. That is, for every R-algebra C and square-zero ideal $N \subseteq C$, every algebra map $E \to C/N$ lifts uniquely to $E \to C$. We separate the two halves: *lifting* (existence) and *unramified* (uniqueness). Flatness is not part of the definition.

For $Q, G \in R[X]$ with Q monic, a *point* of (Q, G) in an R-algebra C is an element $a \in C$ with $Q(a) = 0$ and $G(a)Q'(a)$ a unit. The algebra

$$S = (R[X]/(Q))[1/(GQ')(a)], \qquad a = X \bmod Q,$$

is characterised by its points: S has a point a, and $h \mapsto h(a)$ is a bijection from algebra maps $S \to C$ to points of (Q, G) in C. An algebra is *standard étale* if it is isomorphic to such an S. A standard étale algebra is unramified: two maps that agree modulo a square-zero ideal send a to roots $a_1, a_2 = a_1 + \varepsilon$ with $\varepsilon^2 = 0$. Taylor's formula gives $0 = Q(a_2) - Q(a_1) = Q'(a_1)\varepsilon$, and $Q'(a_1)$ is a unit, so $\varepsilon = 0$.

R is *residually discrete local* with maximal ideal $\mathfrak{m}$: every $a \in R$ is in $\mathfrak{m}$ or a unit, and $1 - a$ is a unit for $a \in \mathfrak{m}$.

Theorem 1.1 (Local structure)**.** *Let E be an étale R-algebra and $\mathfrak{q}$ a prime of E containing $\mathfrak{m}E$, with decidable membership. There is $s' \in E \setminus \mathfrak{q}$ such that $E[1/s']$ is standard étale over R.*

The proof has two parts. Sections 2–6 build a monic Q and a point y of (Q, G) in a localisation of E. This part uses only that E is finitely presented and that its special fibre is finite étale. Section 7 then uses that E is étale. Section 8 replaces this last part when E is flat and unramified instead.

2. The two inputs

The proof uses two theorems as inputs.

Zariski's Main Theorem [1, 3]. Let $x_1, \ldots, x_n$ generate E, and let $E/\mathfrak{m}E$ be finite over $k = R/\mathfrak{m}$. Then there is $s \in 1 + \mathfrak{m}E$ such that $B = R[s, sx_1, \ldots, sx_n] \subseteq E$ is a finite R-module, with generators $b_1, \ldots, b_r$; and $B[1/s] = E[1/s]$. Only the following consequences are used: $s \notin \mathfrak{q}$, because $\mathfrak{m}E \subseteq \mathfrak{q}$ and $1 \notin \mathfrak{q}$; and every $z \in E$ satisfies $s^k z \in B$ for some k, since z is a polynomial in the x_i. The proof does not use that $B \to E$ is injective, so we write $j : B \to E$ only where the distinction

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matters. The coordinates of an element of B in the b_i, and the k with $s^k z \in B$, are known to exist but are not given by a function. At each stage the proof uses finitely many of them, and its conclusion is a proposition, so this is enough (Section 10).

The field case [5, 4]. $(E/\mathfrak{m}E)[1/s]$ is a finite product of monogenic finite étale factors $k[X]/(P_i)$, $k = R/\mathfrak{m}$. Only the following consequences are used. There are $\varepsilon_i, x_i \in B$ and $P_i \in R[X]$ such that, for some N:

- $s^N(\sum_i \varepsilon_i - 1) \in \mathfrak{m}E$;
- $s^N \varepsilon_i P_i(x_i) \in \mathfrak{m}E$ (root);
- $s^N \varepsilon_i (P_i'(x_i) w_i - 1) \in \mathfrak{m}E$ for some w_i (separable);
- for every $b \in B$ there is $C \in R[X]$ with $s^N \varepsilon_i (b - C(x_i)) \in \mathfrak{m}B$ (generation).

Orthogonality and idempotency of the ε_i are not used, nor is monicity of P_i.

Lemma 2.1 (Choice of a factor)**.** *Some ε_i is outside $\mathfrak{q}$. With $s' = s\varepsilon_i$, all the data above hold for s' and the single factor (x_i, P_i).*

Proof. Decide $\varepsilon_i \in \mathfrak{q}$ for each i. If all are in $\mathfrak{q}$, so is $\sum \varepsilon_i$, hence so is s^N, since $s^N(\sum \varepsilon_i - 1) \in \mathfrak{m}E \subseteq \mathfrak{q}$. But $s \notin \mathfrak{q}$ and $\mathfrak{q}$ is prime. For the refinement use $(s\varepsilon)^{N+1} w = (s\varepsilon^N)\big(s^N(\varepsilon w)\big)$ and $(s\varepsilon)^k z = \varepsilon^k (s^k z)$. □

From now on there is one factor: $s \notin \mathfrak{q}$, $x \in B$ and P with

$$s^N P(x) \in \mathfrak{m}E, \qquad s^N(P'(x)w - 1) \in \mathfrak{m}E, \qquad \forall b\ \exists C:\ s^N(b - C(x)) \in \mathfrak{m}B. \tag{2.1}$$

We may assume $x \notin \mathfrak{q}$: otherwise $x - 1 \notin \mathfrak{q}$, and we replace x by $x - 1$ and P by $P(X + 1)$.

3. An idempotent and the determinant lemma

Fix an ideal I of B containing $\mathfrak{m}B$.

Lemma 3.1 (Residual idempotent)**.** *Suppose $\sum_i c_i t^i \in I$ for some $t \in B$, where some c_i is a unit. Then there are v and $N \geq 1$ with $t^N(1 - tv) \in I$. Put $e = (tv)^N$. Then $e^2 \equiv e$, $t^N(1 - e) \in I$, and $e \in tB$. Moreover $t^K w \in I$ implies $ew \in I$.*

Proof. Using decidability of $\mathfrak{m}$, find the first coefficient c_m that is a unit, and divide by it. The coefficients below m are in $\mathfrak{m}$, so this gives $t^m(1 - tv) \in I$. Take $N = m + 1$. From $t^N \equiv t^N(tv)$ we get $t^N \equiv t^N(tv)^N$, and multiplying by v^N gives $e \equiv e^2$. □

B is finite, so s satisfies a monic equation (determinant trick on $b_1, \ldots, b_r$). Lemma 3.1 with $I = \mathfrak{m}B$ and $t = s$ gives an idempotent e modulo $\mathfrak{m}B$. Put $y = xe$. Then $y \notin \mathfrak{q}$: $x \notin \mathfrak{q}$, and $e \notin \mathfrak{q}$ because $s^N(1 - e) \in \mathfrak{q}$. For $b \in B$, by (2.1) and idempotency, modulo $\mathfrak{m}B$,

$$yb = xeb \equiv xe\,C(x) \equiv xe\,C(xe) = (XC)(y),$$

so $yB \subseteq R[y] + \mathfrak{m}B$. In coordinates, $yb_j = c_j + \sum_l \mu_{jl} b_l$ with $c_j \in R[y]$ and $\mu_{jl} \in \mathfrak{m}$.

Lemma 3.2 (Determinant lemma)**.** *Let $p = \det(y\delta_{jl} - \mu_{jl})$. Then $pb_j \in R[y]$ for all j, hence $pB \subseteq R[y]$. For every ideal $J \supseteq \mathfrak{m}B$, $p \equiv y^r \pmod{J}$; in particular $p \notin \mathfrak{q}$.*

Proof. Multiply the system $(yI - \mu)b = c$ by the adjugate. The congruence holds because $\mu \equiv 0$ modulo J, and $y \notin \mathfrak{q}$ with $\mathfrak{q}$ prime. □

Put $g = ps \notin \mathfrak{q}$ and choose $G \in R[X]$ with $G(y) = g$. Then $gB \subseteq R[y]$, and every $z \in E$ satisfies $g^k z = r(y)$ for some k and $r \in R[X]$: if $s^k z = b$, then $g^{k+1} z = p^k \cdot p(bs) \in R[y]$. We call this property *Full*.

4. The special fibre, read in E

Instead of working in E_g we work in E with the ideal

$$J = \{w \in E \mid g^k w \in \mathfrak{m}E \text{ for some } k\},$$

which is $\mathfrak{m}E_g \cap E$ read inside E. Then $J \subseteq \mathfrak{q}$. We take $H = P$ (the note uses $P/\gcd(P, X)$), and $W = G^n H$ for a suitable n.

Lemma 4.1. (1) *$H(y) \in J$;*
(2) *$H'(y)$ is a unit modulo J;*
(3) *g is a unit modulo J;*
(4) *$W(y) = \sum_j \mu_j R_j(y)$ with $\mu_j \in \mathfrak{m}$ and $R_j \in R[X]$.*

Proof. Since $s^N(1-e) \in \mathfrak{m}B$ and g is a multiple of s, we get $1 - e \in J$, so $y \equiv x$ modulo J. Hence (1) and (2) follow from (2.1); for (2), the congruence is carried through the dual numbers, where P' is the ε-component of $P(x + \varepsilon)$. For (3), g satisfies a monic equation, and Lemma 3.1 with $t = g$ gives $g^N(1 - gv) \in \mathfrak{m}B$. For (4), $g^k H(y) \in \mathfrak{m}E$. Write the elements of $\mathfrak{m}E$ involved as $\sum \mu_j z_j$, and clear their denominators by Full, one generator at a time. Both sides then lie in $R[y]$. □

5. The loop

Write $[c]_e = \sum_{i<e} c_i X^i$. A monic relation of degree e for y is $y^e = [a]_e(y)$, and $Q = X^e - [a]_e$. Let $V_e(\mu) = \sum_{i<e} \mu_i y^i$.

Proposition 5.1. *Let Q_0 be a monic relation for y and suppose $W(y) = \sum_j \mu_j R_j(y)$ with $\mu_j \in \mathfrak{m}$. There are a monic relation Q with $Q(y) = 0$, a polynomial U, and $c_0, \ldots, c_{e-1} \in \mathfrak{m}$ with*

$$W = UQ + [c]_e \quad \text{in } R[X].$$

Thus $\bar{Q}$ divides $\bar{W}$ in $k[X]$.

Proof. (a) *Reduction.* Every F is $UQ + [c]_e$ (Horner's scheme, using $X^e = Q + [a]_e$).

(b) *The $\mathfrak{m}$-part.* The set of $V_e(\mu)$ with all $\mu_i \in \mathfrak{m}$ is an $R[y]$-submodule; for multiplication by y, use $y^e = [a]_e(y)$. So $W(y)$ belongs to it.

(c) *One step.* Reduce $W = UQ + [c]_e$. Scan $c_{e-1}, \ldots, c_0$ with the decidability of $\mathfrak{m}$. If all are in $\mathfrak{m}$, stop. Otherwise let $d < e$ be the top index with c_d a unit. Then $D = X^d + c_d^{-1}[c]_d$ is monic, and by (b), $D(y)$ lies in the $\mathfrak{m}$-part. Hence every y^j is $V_d(\beta_j)$ plus an element of the $\mathfrak{m}$-part. For $j < e$ this is a system $(1 - \mu)v = n$ with $v_j = y^j$ and $\mu \equiv 0$ modulo $\mathfrak{m}$. The determinant $\det(1 - \mu) \equiv 1$ is a unit because R is local, and the adjugate gives $y^j \in V_d(R)$. For $j = d$ this is a new monic relation of degree $d < e$; the same formulas move $W(y)$ into the new $\mathfrak{m}$-part.

The degree strictly decreases, so the loop terminates. □

The note takes $D_l = \gcd(\bar{Q}_l, \bar{G}^n \bar{H})$. Here D_l is the remainder of W modulo Q_l, made monic at its top unit coefficient; the loop thereby performs Euclid's algorithm itself.

6. The derivative

Lemma 6.1. *$U(y)Q'(y) \equiv G(y)^n H'(y) \pmod{J}$. Hence $Q'(y)$ is a unit modulo J, and $Q'(y) \notin \mathfrak{q}$.*

Proof. Evaluate $W = UQ + [c]_e$ at $y + \varepsilon$ in $E[\varepsilon]/(\varepsilon^2)$ and compare ε-components. We have $Q(y) = 0$, the derivative of $[c]_e$ has coefficients in $\mathfrak{m}$, and $W' = nG^{n-1}G'H + G^n H'$ with $H(y) \in J$. The right side is a unit modulo J by Lemma 4.1, and $J \subseteq \mathfrak{q}$. □

Neither $\bar{H} \mid \bar{Q}$ nor the factor $D = \bar{Q}/\bar{H}$ of the note is needed.
Put $g_1 = g\,Q'(y) \notin \mathfrak{q}$, $T = E[1/g_1]$, and

$$S = (R[X]/(Q))[1/(GQ')(a)].$$

The element y is a point of (Q, G) in T, so it defines $u : S \to T$ with $u(a) = y$.

7. The étale case

Lemma 7.1. (1) *u is surjective.*
(2) *Let $x_1, \dots, x_v$, $f_1, \dots, f_w$ be a presentation of E, with $x_i = \sigma_i(y)/g^k$ by Full and $y = Y(x)$. Then $\ker u$ is generated by $f_j(\sigma(a))$ and $a - Y(\sigma(a))$, where $\sigma_i(a)$ means $\sigma_i(a)/G(a)^k$.*
(3) *T has the lifting property.*

Proof. (1) By Full, $z \in E$ is $r(y)/g^k = u(r(a)G(a)^{-k})$, and $1/g_1 = u(1/(GQ')(a))$. (2) Let K be the displayed ideal. The map $E \to S/K$, $x_i \mapsto \sigma_i(a)$, kills the f_j and sends y to a and g_1 to a unit. So it extends to $\psi : T \to S/K$. Then ψu is the projection, since both send a to a. Hence $u(z) = 0$ implies $z \in K$. (3) A map $T \to C/N$ gives $E \to C/N$, which lifts to $E \to C$ because E is étale. The image of g_1 is a unit modulo the square-zero ideal N, hence a unit, so the lift extends to T. □

Proposition 7.2. *Let $I = \ker u$. Then $I = I^2$, and there is an idempotent f with $I = (1 - f)$. Consequently $T \cong S[1/f]$ is standard étale, for (Q, GF) where $f\,(GQ')(a)^n = F(a)$.*

Proof. Let $C = S/I^2$ and let N be the image of I, which is square-zero. Then $C/N = T$, and lifting gives $\sigma : T \to C$. The projection $S \to C$ and σu agree modulo N. Since S is unramified they are equal, so I maps to 0, that is, $I \subseteq I^2$. With generators $i_j = \sum_l a_{jl} i_l$, $a_{jl} \in I$, put $f = \det(1 - a)$. Then $fI = 0$ and $1 - f \in I$, so $f^2 = f$ and $I = (1 - f)$. Maps out of S that kill I are exactly those inverting f, equivalently $F(a)$. So T has the universal property of $S[1/f]$, which is standard for (Q, GF). □

Proof of Theorem 1.1. Take $s' = g_1 \notin \mathfrak{q}$. Then $E[1/s'] = T$ is standard étale by Proposition 7.2. □

8. The flat and unramified case

An R-module M is *flat* if every relation is explained [5]: if $\sum_i a_i m_i = 0$ with $a_i \in R$ and $m_i \in M$, there are $c_{ij} \in R$ and $n_j \in M$ with $m_i = \sum_j c_{ij} n_j$ and $\sum_i a_i c_{ij} = 0$ for all j.

Theorem 8.1. *Let E be a finitely presented, flat and unramified R-algebra and $\mathfrak{q}$ as in Theorem 1.1. There is $s' \in E \setminus \mathfrak{q}$ such that $E[1/s']$ is standard étale over R, hence étale.*

Sections 2–6 and Lemma 7.1(1),(2) do not use the lifting property. Unramifiedness enters only through the field case. They give $u : S \to T = E[1/g_1]$, surjective, with finitely generated kernel I. As a localisation of E, T is flat over R.

Proposition 8.2 (Transfer)**.** *Let S be standard étale for (Q, G), with point a, and let $\varphi : S \to T$ be a map of R-algebras with T flat over R. Then T is flat over S.*

Proof. Put $D = (T[X]/(Q))[1/(GQ')(X)]$, that is $S \otimes_R T$. It is flat over S, by flat base change through polynomials, a principal quotient and a localisation. There are maps $\lambda : S \to D$, $a \mapsto X$, $\rho : T \to D$ and $\mu : D \to T$, $X \mapsto \varphi(a)$, with $\mu\rho = \mathrm{id}$. Write $\alpha = \lambda(a)$ and $\beta = \rho(\varphi(a))$. Both are roots of Q in D, and $Q'(\alpha)$ is a unit. By synthetic division $Q = (X - \alpha)H$ in $D[X]$, so $H(\alpha) = Q'(\alpha)$ and $(\beta - \alpha)H(\beta) = 0$. Put $e = H(\beta)/H(\alpha)$. Then $e\alpha = e\beta$, and $\mu(e) = 1$ because μ sends α and β to $\varphi(a)$. In $D/\mathrm{Ann}(e)$ the images of α and β are equal. So the two maps $S \to D/\mathrm{Ann}(e)$ induced by λ and $\rho\varphi$ agree at a, hence are equal by the universal property of S.

Thus $e\lambda(x) = e\rho(\varphi(x))$ for every $x \in S$. Therefore $t \mapsto e\rho(t)$ is S-linear and a section of μ. So T is an S-module retract of D, and flat. The element e need not be idempotent. □

Lemma 8.3 (Flat quotient)**.** *If $u : S \to T$ is surjective and T is flat over S, then every $x \in \ker u$ satisfies $x = xb$ for some $b \in \ker u$. Hence $I = I^2$.*

Proof. The relation $x \cdot 1 = 0$ in T is explained: $1 = \sum_j u(c_j)t_j$ with $xc_j = 0$. Choose t'_j with $u(t'_j) = t_j$, and put $b = 1 - \sum_j c_j t'_j$. Then $u(b) = 0$ and $xb = x$. □

Proof of Theorem 8.1. By Proposition 8.2 and Lemma 8.3, $I = I^2$. The end of the proof of Proposition 7.2 gives an idempotent f with $I = (1 - f)$, so $T \cong S[1/f]$ is standard étale. Take $s' = g_1$. □

A standard étale algebra is étale. Uniqueness of lifts is the unramifiedness of Section 1. For existence, let $a \in C$ reduce to a point modulo N, with $N^2 = 0$, and let w be an inverse of $Q'(a)$ modulo N. Then $b = a - Q(a)w$ satisfies $Q(b) = 0$ by Taylor's formula, since $Q(a) \in N$. Also $(GQ')(b)$ is a unit, since it is a unit modulo N.

The note proceeds differently. It injects $I/\mathfrak{m}I$ into $S/\mathfrak{m}S$ by flatness, uses the special-fibre isomorphism $S/\mathfrak{m}S \cong T/\mathfrak{m}T$ to get $I = \mathfrak{m}I$, and then applies Nakayama's lemma, which needs a further localisation. Here neither the special-fibre isomorphism nor a further localisation is needed. Flatness cannot be omitted: $\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ is surjective, finitely presented and unramified, and its kernel (2) is not idempotent.

9. Examples

Localisation. If $t \in \mathfrak{m}$, then $E = R[X]/(1 - Xt)$ is $(R[Y]/(Y))[1/t]$: points are x with $xt = 1$, and these correspond to t being a unit. Its special fibre is zero.

Adding a simple root. Let $P = aX^2 + bX + c$ and $E = (R[X]/(P))[1/P'(x)]$, with $D = P'(x) = 2ax + b$. Put $y = ax + b$, $s = yD$, $F = Y^2 - bY + ac$ and $G = Y(2Y - b)$. Then $x \mapsto -c/Y$ and $Y \mapsto y$ identify $E[1/s]$ with $(R[Y]/(F))[1/G]$, which is standard since $F' = 2Y - b$. Where a is a unit, P/a is monic, which gives a second chart. The charts cover E, by $1 = ax/D + s/D^2$. Inverting Y is necessary: for $R = \mathbb{Z}$ and $P = 2X^2 + X$, $Y = 0$ is a point of $(R[Y]/(F))[1/F']$ but not of $E[1/s]$.

10. The formalisation

10.1. **Where and how.** There are three developments, checked by Unit (`https://github.com/coquand/unit`):

- `unit/sag/etale-local`: Theorem 1.1 with ZMT as a hypothesis, in the weak form of Section 2 (`TheoremLocal`);
- `unit/sag/etale-local-zmt`: this development and `unit/sag/zmt-general` (the proof of ZMT [3]) in one tree; the shared library files are the same files. The module `TheoremLocalZMT` discharges the ZMT hypothesis with `FinalZMTGeneral.Theorem`;
- `unit/sag/etale-flat-unramified`: Theorem 8.1, with the same weak ZMT and field case as hypotheses (`FlatTheoremLocal`).

All three are parametrised by the segment `Theory` of `sag/`: univalence (`univ`) and replacement (`images`, `surjections`), with function extensionality and propositional univalence derived. No new axiom is introduced. On 30 October 2026 fresh checks gave: `etale-local`, 3985 declarations in 89 modules, 0 failed, 1.30 s; `etale-local-zmt`, 21057 declarations in 394 modules, 0 failed, 14.27 s. On 31 October 2026: `etale-flat-unramified`, 7076 declarations in 162 modules, 0 failed, 2.43 s. Each file starts with its informal proof.

10.2. **The statement.** The final theorem is

`TheoremLocalZMT.Over[...].FieldCase[field].theorem :  || Conclusion ||`

with `Conclusion` from `Local.Statement`:

$$(s' : E) \times \neg\mathfrak{q}(s') \times (T,\ i : E \to T) \times \texttt{IsLocalisation}\ T\ i\ s' \times (Q\ G) \times \texttt{Standard}_{Q,G}\ T.$$

`IsLocalisation` is the universal property of $E[1/s']$: $i(s')$ is a unit, and restriction along i is an equivalence from maps $T \to C$ to maps $E \to C$ inverting s'. `Standard` is the universal property of $(R[X]/(Q))[1/(GQ')(a)]$ by its points (Section 1). Thus isomorphisms of the note are stated by points, which identifies the algebras by Yoneda.

The parameters of `TheoremLocalZMT.Over` are:

- R with $\mathfrak{m}$ given by its closure properties, membership a proposition (`m-prop`), decidability `dec` and locality `rad`;
- E, the prime $\mathfrak{q}$ (closure, `q-prime`, `q-proper`, `q-m`: $\mathfrak{m}E \subseteq \mathfrak{q}$) with decidable membership `q-dec`;
- `etale`: E formally étale (`Lift`, `Unique`) together with a presentation; its generators are the x_i given to ZMT;
- `fo`: merely, $E/\mathfrak{m}E$ is a finite $R/\mathfrak{m}$-module (the hypothesis of ZMT).

The parameter of `FieldCase` is `field`: for every output c of ZMT (an s with $s-1 \in \mathfrak{m}E$, finiteness of $B = R[s, sx_i]$ and $B[1/s] = E[1/s]$), merely a `FieldData`: the ε_i, the identity `sum-one`, and for each factor a `Piece`: P with the three facts (2.1) multiplied by ε_i.

The intermediate theorem `TheoremLocal.Over[...].theorem-of` has, in place of `etale`'s generators and `fo`, the weak ZMT of Section 2: B, j, generators `bs`; merely the coordinates of each element (`coord`); s with merely $j(s) - 1 \in \mathfrak{m}E$ (`s-one`); and, for each z, merely k and b with $j(s)^k z = j(b)$ (`zmt`).

For Theorem 8.1 the statement is

`FlatTheoremLocal.Over[...].theorem fd :  || Conclusion ||`

with `fd` merely a `FieldData`. Its parameters are those of `TheoremLocal.Over`, with `etale` replaced by a presentation of E and `flat`: E is flat over R in the sense above. `Conclusion`, from `FlatLocalStatement`, is the conclusion above together with `Etale` T. The variant `from-unramified` takes `Unique` E and a principle `Unique` $E \to \|$`FieldData`$\|$, which is how the field case uses unramifiedness.

10.3. **Correspondence.**

Mathematics	Unit module
Definitions, Section 1	`FormallyEtale` (`SquareZero`, `Lift`, `Unique`, `Etale`), `StandardEtale` (`Points`, `Standard`, Taylor in dual numbers, `unramified`)
Choice of a factor	`TheoremLocal` (`find`, `Refine`, `FieldCase.At`)
x versus $x-1$, $y = xe$, system	`Introduction` (steps 0–3), `HornerValues`
Lemma 3.1	`ResidualIdempotent` (over a discrete field), `IdempotentModM` (modulo an ideal $\supseteq \mathfrak{m}$)
Lemma 3.2	`DeterminantLemma` (`System`, `Congruence`, `Prime.p-out`), on `sag/exterior`
Full, Lemma 4.1	`Introduction` (step 4, `Jsat`), `ClearDenominators`, `IdealGenerated`, `LoopInputs`
Proposition 5.1	`PowerSums`, `MonicReduce` (a), `MPart` (b), `LoopStep`, `LoopNakayama` (c), `Loop`
Lemma 6.1	`DerivativeStep` (`Qprime`), `LoopQprime`
S	`StandardAlgebra` (built from a presentation and `Localization`)

Mathematics	Unit module
Lemma 7.1	`Local`, steps (b)–(e)
Proposition 7.2	`EtaleCase` (`Surjection`, `Square`, `Localised`), `IdempotentIdeal`
Theorem 1.1	`Local.Field.theorem`, `TheoremLocal`
ZMT plugged in	`TheoremLocalZMT` (`At: coord`, `s-one`, `zmt`), on `FinalZMTGeneral.Theorem`, `CertificateSubalgebra`, `CertificateLocalisation.Iso.cleared`
Section 8: flatness	`FlatRelations` (explained relations), `FlatSystems`, `FlatPolynomialBaseChange`, `FlatPrincipalQuotient`, `FlatPowerLocalization`
Proposition 8.2	`FlatStandardChart` (D), `FlatRootFactor` (synthetic division), `FlatSeparableRetraction` (e), `FlatRetraction`, `FlatStandardTransfer`
Lemma 8.3	`FlatQuotient`
Theorem 8.1	`FlatBaseStandardQuotient`, `FlatLocal`, `FlatTheoremLocal`
Standard implies étale	`StandardEtaleLift`
Examples	`LocalisationExample`, `SimpleRootExample` (with `cover` and `Necessity`)

10.4. **Changes with respect to the note.** The proof above already incorporates these changes. We list them for a reader of `zmt2-revised.tex`.

(1) *Hypotheses used.* Of ZMT only $s \in 1+\mathfrak{m}E$ and $s^k z \in B$ are used, and $B \to E$ need not be injective. Of the field case only the identity $\sum \varepsilon_i \equiv 1$ and the three facts of each factor are used. The factor and the refinement $s' = s\varepsilon_i$ are chosen in the proof, using decidability of $\mathfrak{q}$.
(2) *Lemma 1.* The hypothesis is any relation with a unit coefficient, and $N = m+1$ always. The lemma is applied modulo $\mathfrak{m}B$ without forming $A = B/\mathfrak{m}B$.
(3) $H = P$. The gcd with X is not needed, because y is a unit modulo J.
(4) *Section 1 in E.* The note's $\mathfrak{m}E_g$ becomes the saturation J in E, so no localisation is constructed for Section 1.
(5) *Loop.* Remainders of W replace $\gcd(\bar{Q}_l, \bar{W})$.
(6) *Derivative.* $Q'(y)$ is a unit modulo J by the ε-component identity, without $\bar{H} \mid \bar{Q}$ and without the special-fibre isomorphism.
(7) *Kernel of u.* The note obtains finite generation from finite presentation. Here explicit generators are constructed from the presentation of E (Lemma 7.1(2)).
(8) *Mere existence.* Lists of coefficients for the polynomials of Lemma 3.2, expressions in generators, and certificates of $I = I^2$ are chosen under truncation. Hence the conclusion is truncated. I^2 must have propositional membership for the quotient ring.
(9) *Finitely many coordinates.* ZMT gives coordinates in B, the exponents k with $s^k z \in B$, and $s - 1 \in \mathfrak{m}E$ only merely. Full is therefore a mere property, and the proof chooses, in stages, the finitely many instances it uses: the coordinates of 1, s and sb_i; then (they depend on e) those of the elements $yb_i - (XC_i)(y)$ of $\mathfrak{m}B$, with coefficients in $\mathfrak{m}$; then (they depend on p) those of gb_i and yb_i; and Full at the generators of E, for the preimages in Lemma 7.1. Denominators are cleared (Lemma 4.1) by the same induction under the truncation.
(10) *The ZMT interface.* The B of ZMT is the image subalgebra $R[s, sx_i]$ of E. A zero generator is added to its module generators, so that there is at least one. The exponent k with $s^k z \in B$ is obtained by clearing the denominators of an expression of z in the x_i, not from the inverse of $B[1/s] \to E[1/s]$.

(11) *The flat case.* The special-fibre comparison and Nakayama's lemma are replaced by Proposition 8.2 and Lemma 8.3. The conclusion is reached at $s' = g_1$, without a further localisation.

10.5. **What remains.**

- *The field case* (structure of finite étale algebras over a discrete field [5, 4]) is a hypothesis, in two parts: $E/\mathfrak{m}E$ is finite (`fo`), and the factor data for the output of ZMT (`field`). The options, including small residue fields, are analysed in `sag/etale-local/FIELD-CASE.md`. In the flat case it enters through `FieldData`, or through the principle `Unique` $E \to \|$`FieldData`$\|$.
- *ZMT in the flat case* is still the weak hypothesis of Section 2; it has not been plugged in. Of `etale`, `TheoremLocalZMT` uses only the presentation of E, which the flat case also has.
- *Not formalised*: the global version (the lemma on changing the base from $R[1/f]$ to R, the finite-character lemma, and the theorem for arbitrary R with the dynamical local–global principle), in both the étale and the flat case; the remark on $R = k[t]_{(t)}$; and the identification $E \cong \mathbb{Z} \times \mathbb{Z}[1/2]$ in the simple-root example.

References

[1] M. Alonso, T. Coquand and H. Lombardi, *Revisiting Zariski Main Theorem from a constructive point of view*, J. Algebra 406 (2014), 46–68.
[2] T. Coquand, *Local structure of étale algebras*, `zmt2-revised.tex`.
[3] *A constructive proof of Zariski's Main Theorem by finite section lifting, and its formalisation in Unit*, `unit/doc/zmt-proof.tex`.
[4] H. Lombardi, *Un théorème décisif*, arXiv:2506.07098, 2025.
[5] H. Lombardi and C. Quitté, *Commutative Algebra. Constructive Methods*, Springer, 2015.